

THE COMMUTE QUESTION

A statistics case study built on real-shaped, deliberately messy survey data. You will not be told which technique to use. You will have to decide.

Course	Introduction to Statistics — Day 1 & 2
Deliverable	A one-page recommendation backed by evidence, plus your cleaning log
Data files	ITI_Statistics_CaseStudy_Data.xlsx / iti_survey_raw.csv
Tools	Excel, Google Sheets, or Python (pandas) — your choice
Working style	Teams of 3–4. Every team member owns at least one section.
Time	Roughly 4–6 hours of work

Read this before you open the file

This case has no single correct answer, but it has many wrong ones.

Two teams can reach different conclusions and both be defensible — **if** they show how they got there. A team that reports a confident number with no reasoning behind it has failed the exercise, even when the number happens to be right.

Statistics is not the arithmetic. The arithmetic is the easy part and your laptop does it. Statistics is deciding **which** number to compute, on **which** rows, and what it is honestly allowed to mean.

1. The situation

ITI is considering a policy change. Several instructors have argued that students who travel a long distance to the branch are being quietly disadvantaged — they arrive tired, they leave early, and their results suffer for reasons that have nothing to do with ability. The proposal on the table is to offer a **remote-attendance option two days per week for students with long commutes**.

The proposal is not free. It costs equipment, coordination, and instructor time. Before committing, the academic director wants evidence rather than anecdote, and has asked your team to look at the data.

The brief you received

Memo from the Academic Director

"We surveyed 260 students across five branches. I want to know three things.

One. Is the commute problem real, and how big is it? I keep hearing 'students travel about an hour'. Is that true?

Two. Does commuting actually hurt performance? If it does not, this policy solves nothing and I will spend the budget elsewhere.

Three. If we run this, who qualifies? Give me a threshold I can defend to the board, and tell me how many students it covers.

I need one page. Keep the workings, I may ask to see them."

What makes this hard

The director asked a causal question ("does commuting hurt performance") and expects a threshold ("who qualifies"). Your data can support one of those better than the other. Part of your job is to answer what the data can answer, and to say clearly where you had to stop.

2. Your data

You have **271 rows** and 15 columns in `raw_data`. Note that 271 is not the number of students surveyed. Finding out why is your first task.

The `data_dictionary` sheet lists every column, its variable type, and what a valid value looks like. Read it before you touch anything. Most of the mistakes made on this case study are made by teams that skipped it.

Part A — Warm-up: classify the variables

Before any analysis, sort all 15 columns into the classification tree. This is not busywork: the variable type determines which statistics are even legal to compute.

Quantitative — Discrete	Quantitative — Continuous	Qualitative — Ordinal	Qualitative — Nominal

Three questions that decide whether you understood this

- a.** `satisfaction_1to5` is stored as the numbers 1 to 5. Excel will happily average it and give you 3.6. Is that number meaningful? What does the gap between "2" and "3" represent, and is it the same size as the gap between "4" and "5"?
- b.** `student_id` is made of digits. Which measures of central tendency can you legitimately compute on it? Which would Excel compute anyway?
- c.** Age is recorded in whole years, so it is discrete here. But age itself is continuous. What did the survey design do to it, and does that matter for your analysis?

3. Part B — The data is dirty. Deal with it.

Real data does not arrive analysis-ready. This file contains at least seven distinct categories of problem. Your task is to find them, decide what to do about each one, and **write down why** in the `cleaning_log` sheet.

The seven categories

#	Category	What it looks like	The question you must answer
1	Duplicate records	The same student appearing more than once — sometimes identical, sometimes with one field changed	Which copy is the truth? Does an exact duplicate mean the same thing as a near-duplicate?
2	Inconsistent categories	"Cairo", "cairo", "CAIRO", "Alexandria", "Mansora", "DS", "M" vs "Male"	How many branches are there really? What happens to your counts if you skip this step?
3	Missing values, many disguises	Blank cells, "NA", "n/a", "missing", "?", "unknown", "-"	Are these all the same thing? Does a blank mean zero, or mean nothing?
4	Impossible values	age = 999, age = -1, sleep_hours = 25, final_score = 150, pets = -2	Are these observations or typos? Delete the value, or delete the whole student?
5	Numbers stored as text	"45 min", "18.5 hours", "78,4" (comma decimal)	Excel treats these as text and silently excludes them from AVERAGE. Did you notice?
6	Genuine extremes	A handful of very long commutes that are perfectly plausible	These are not errors. What is the cost of deleting them? What is the cost of keeping them?
7	Ambiguous answers	"None", "none", "No sport", "-"	Does "None" mean the student plays no sport, or that they did not answer? These are different.

The distinction that matters most on this whole case study

An error is a value that could not have happened. An outlier is a value that did happen and is unusual.

A sleep value of 25 hours is an error — the night is not that long. A commute of 240 minutes is an outlier — miserable, but entirely possible for someone travelling in from outside the city.

You remove errors. You **keep** outliers and choose statistics that can cope with them. Teams that delete every uncomfortable number end up with a clean dataset that describes a population which does not exist.

Deliverable for Part B

A completed `cleaning_log` with one row per decision, and a single sentence at the top: *"We started with 271 rows and finished with N rows because ____."* Your N and your reasoning both get marked.

4. Part C — Describe what you have

Now answer the director's first question: **is the commute problem real, and how big is it?**

C1. The headline number

Compute, for `commute_minutes` across all cleaned rows: the mean, the median, the mode, the range, the standard deviation, Q1, Q3 and the IQR.

Then stop and look at what you produced

The mean and the median are noticeably different. That gap is not noise, and it is not a mistake in your arithmetic. It is the single most informative thing in this dataset.

Which one is bigger, and what does that tell you about the shape of the distribution? Sketch what the histogram must look like before you plot it, then plot it and check whether you were right.

The director said he keeps hearing "students travel about an hour". Using your numbers, is that claim true, false, or true-but-misleading? Defend your answer in two sentences. This is the question, more than any other, that separates a team that computed statistics from a team that understood them.

C2. Which average should you report?

You now have three measures of centre that disagree. You must report one to the director as "the typical commute".

1. Which do you choose, and why? Reference the sensitivity of each measure to extreme values.
2. Suppose a rival team reports the mean and you report the median. Both are correct arithmetic. Whose number better serves the decision being made, and can you name a situation where the other choice would be the right one?
3. The mode of a continuous variable is close to useless here. Explain why — and then explain why the mode is the **only** option you have for transport.

C3. Spread, and why the average alone is a lie

1. Report the standard deviation and the IQR for commute time. They tell different stories. Why?
2. Compute the coefficient of variation ($CV = 100 \times s / \text{mean}$) for commute time and for final score. Which variable is more spread out **relative to its own size**? Note that you could not have compared these two directly using the standard deviation alone — the units are different.
3. Build the $1.5 \times \text{IQR}$ fences for commute time. How many observations fall outside them? Look at those specific students. Do you believe them?

C4. Cut the data by group

Produce a summary table of commute time **by branch**: count, mean, median, and standard deviation for each of the five branches.

1. One branch is dragging the overall average upward. Which, and by how much?
2. Recompute the overall mean commute with that branch excluded. How much does the headline number move? What does that tell you about whether a single national policy is the right shape?
3. One branch has a much smaller standard deviation than the others. What does a small dispersion mean in plain language for the students at that branch?

A trap that has caught every cohort so far

Your five branches do not have equal numbers of students. When you combine them, the biggest branch pulls the overall average toward itself.

So: is the "average ITI commute" the average of the five branch averages, or the average of all students? These give different answers. **Which one did you compute, and which one did the director actually ask for?**

5. Part D — Does commuting actually hurt performance?

This is the director's second question and the one the whole policy rests on.

D1. Look before you calculate

Draw a scatter plot: `commute_minutes` on the x-axis, `final_score` on the y-axis. Do this **before** computing any correlation coefficient.

1. Describe what you see in words. Is there a visible relationship? Is it strong, weak, or absent?
2. Now compute the correlation between the two. Does the number match your impression of the plot?
3. Also compute the correlation between `study_hours_week` and `final_score`, and between `sleep_hours` and `final_score`. Rank all three relationships by strength.

What you are about to find, and what to do about it

One of those three relationships is far stronger than the other two. It is probably not the one the policy is about.

This is the most uncomfortable moment in the case study, and the most realistic one. You were asked to evaluate a commute policy and the data is pointing somewhere else. You have three options: bury it, overstate it, or report it honestly and let the director decide.

Report it. Being the person who says "the thing you asked about is not the main driver, here is what is" is the entire value of an analyst. Anyone can confirm what they were sent to confirm.

D2. The causation question

Suppose you find a negative correlation between commute time and final score, even a weak one.

1. Does that establish that long commutes **cause** lower scores? State your answer and defend it.
2. Propose two alternative explanations that would produce the same correlation without commuting causing anything. Think about who lives far from a branch, and what else is true about them.
3. What data would you need — that you do not have — to move from "these two things move together" to "one of them causes the other"? Describe the study you would run.

D3. Compare two groups properly

Split students into "long commute" and "short commute" using a threshold you justify — the median, Q3, or the upper IQR fence are all defensible starting points. Then compare final scores across the two groups.

1. Report the mean, median and standard deviation of final score for each group.
2. How large is the gap between the groups compared with the spread **within** each group? A gap of two marks means very little when the standard deviation inside each group is ten.
3. You have a sample, not the full ITI population. If you repeated this survey next term with different students, would you expect exactly the same gap? What is the name for the difference between your sample result and the truth about the population?

D4. Choose the right graph

For each of the following, name the graph you would use and defend it in one sentence. Use the vocabulary from the course: univariate or bivariate, quantitative or qualitative.

You want to show...	Graph	Why
The shape of the commute-time distribution across all students		
That commute times are heavily concentrated at the low end with a long tail		
A side-by-side comparison of commute time across all five branches		
The relationship between study hours and final score		
The breakdown of preferred transport modes		
How many students fall into each satisfaction level, 1 to 5		

One of these six is a trap: the most obvious chart type is the wrong choice because of the variable type involved. Identify which, and say what you would use instead.

6. Part E — Can you trust this sample at all?

Everything above assumed the 260 students in your file fairly represent ITI. Time to test that.

E1. How the sample was actually collected

What the survey team told you afterwards

"We sent the form out on the internal portal and left it open for a week. Anyone who wanted to answer, answered. We got about 260 back.

The Cairo branch responded most enthusiastically — it is our biggest branch and the coordinator there pushed it in the group chat.

A few students never filled it in. We assumed they were the ones who did not care."

1. Name this sampling method. Is it simple random, stratified, cluster, multistage — or none of them?
2. The course defines an unbiased sample as one where every member of the population has an equal chance of being selected. Did every ITI student have an equal chance here? Point to the exact sentence in the memo that breaks the requirement.
3. The survey ran on the internal portal. Which students are systematically less likely to have responded? Look at the `has_laptop` column and think about what you have just realised.
4. The last sentence of the memo contains an assumption that could be badly wrong. State it, and explain how it could bend your results in a specific direction.

E2. Design it properly

1. You are asked to rerun this survey next term and get it right. The population is 2,000 students across the five branches, in five tracks. Design the sampling strategy. Name the technique, define your strata or clusters, and say how you would draw within them.
2. Compare stratified sampling against cluster sampling for this specific problem. One of them is clearly better here. Which, and why does the structure of ITI make it better?
3. A colleague suggests surveying every student — no sampling at all. Give two concrete reasons from the course why sampling is preferred, and one situation where the census would be the right call.

E3. The missing data problem, revisited

Go back to your missing values. You probably dropped or ignored them in Part B.

1. For the column with the most missing values, check whether missingness is spread evenly across branches and tracks, or concentrated in one group.
2. If it is concentrated, dropping those rows does not just shrink your sample — it changes who is in it. Explain the consequence for your conclusions in plain language.

7. Part F — The recommendation

One page. Written for the academic director, who is not a statistician and will read it in about four minutes.

Section	What goes in it	Length
The answer	Your recommendation: run the policy, do not run it, or run a modified version. State it in the first sentence. Do not build up to it.	2–3 lines
The evidence	The three or four numbers that drive your recommendation. Say which measure of centre you used and why, given the shape of the distribution.	5–6 lines
Who qualifies	Your proposed commute threshold, the number and percentage of students it covers, and why you drew the line there rather than somewhere else.	3–4 lines
What we cannot conclude	The limits. Correlation versus causation. Sampling bias. Missing data. Anything you had to assume.	4–5 lines
What we should do next	One concrete proposal to get better evidence.	2–3 lines

The section that carries the most marks is "What we cannot conclude"

This is deliberate and it is not a trick.

Anyone who has watched a tutorial can produce a mean and a bar chart. What makes someone worth listening to is knowing the boundary of their own evidence — being able to say "this number is solid, that one is shaky, and this question my data simply cannot answer."

A recommendation with honest limits is far more useful, and far more likely to be acted on, than a confident one that quietly falls apart the first time someone senior pushes on it.

Also submit

- Your cleaning log, completed.
- Your working file, with the raw sheet untouched.
- Four charts, each with a title, labelled axes, and units.
- A summary table of commute time and final score by branch.

8. How you will be assessed

Weight	Criterion	What full marks looks like
25%	Data cleaning judgement	Every decision logged with a rule, not a feeling. Errors removed, genuine outliers kept and flagged. The distinction is applied consistently, not case by case.
20%	Correct choice of statistic	Median chosen over mean where the distribution demands it, and the choice explained. No means computed on nominal data. Ordinal data handled honestly.
20%	Interpretation	Numbers are translated into sentences a non-statistician can act on. Shape is read off the mean–median gap. Spread is discussed, not just centre.
20%	Intellectual honesty	Causation not claimed from correlation. Sampling bias identified and its direction reasoned about. The inconvenient finding in Part D is reported rather than buried.
15%	Communication	Charts are readable and correctly chosen for the variable type. The recommendation is one page and leads with the answer.

A last word

Every number in this dataset is fictional. The mess is not.

Every problem you are about to hit — the duplicated rows, the "45 min" that Excel refuses to average, the branch label spelled three ways, the finding that answers a question nobody asked — is something you will meet again in your first month of real work.

Learning to compute a standard deviation takes an afternoon. Learning to be the person in the room who asks "wait, what does this blank cell actually mean?" takes longer, and is worth considerably more.